MIL-S-36298(DM)
7 October 1964

MILITARY SPECIFICATION

SIGMOIDOSCOPE, ELECTRIC, DISTAL AND PROXIMAL LIGHTING, 63 FR

1. SCOPE

1.1 This specification covers a modified Vernon David proctosigmoidoscope of the pneumatic type, having provisions for both proximal and distal lighting, and suitable for use in the examination and treatment of the human rectum and the sigmoid flexure.

FSC 6515

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-P-116	Preservation, Methods of.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Federal

Fed. Test Method Std. No. 601	Rubber: Sampling and Testing.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Medical Supply Center

1927	Sigmoidoscope, Electric, Distal and Proximal Lighting, 63 Fr.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Material.

3.1.1 Metal parts. Metal parts shall be fabricated of brass, and shall comply with plating and corrosion tests requirements of class 4, composition N of GG-I-526.

3.1.2 Rubber parts. Rubber parts, except the light cord, shall be manufactured from a natural rubber compound (see 3.2.8).

3.1.3 Other material. All other material used shall be of high quality, suitable for the purpose intended.

3.2 Style, design, and dimensions. The proctosigmoidoscope shall consist essentially of a proctosigmoidoscope tube, obturator, light carrier and window attachment, light carrier extenion rod, window, electric connector cord, lamps and sleeve, insufflator complete with connector, and carrying case. Style, design, and dimensions of parts shall be as shown in Drawing 1927, and as specified herein.

3.2.1 Proctosigmoidoscope tube. The proctosigmoidoscope tube shall be well constructed with the joint at the proximal end silver soldered. All edges shall be smoothly rounded and shall be without sharpness. The tube shall be graduated as shown in Drawing 1927, with each division completely encircling the tube, or the circle may have a gap of approximately 1/8 inch. The appropriate number for each fifth division shall be shown on two sides (opposite sides) of the tube. The inside of the tube shall be cleanly plated but shall have a dull nonreflective finish. The proximal end of the tube shall be tapered to mate with and provide an airtight fit with the light carrier and window attachment. The obturator shall completely occlude the lumen of the distal end of the tube. The distal end of the obturator shall be provided with a vent as shown in Drawing 1927.

3.2.2 Light carrier and window attachment. The light carrier and window attachment shall have an internal taper at the proximal end to mate with and provide an airtight fit with the window. The distal end shall have an external taper to mate with and provide an airtight fit with the proctosigmoidoscope tube, shall be provided with means for making connection with the electrical connector cord, and shall also contain the receptacle for the proximal light and the light carrier extension for distal lighting. The light carrier and window attachment shall also be provided with means for attaching the insufflator, which shall have an internal taper to mate with and provide an airtight fit with the insufflator connector.

3.2.3 Light carrier extension rod. The light carrier extension rod shall be threaded at the proximal end to mate with the receptacle in the light carrier and window attachment, and at the distal end to mate with the distal lighting bulb, and thus provide electrical connection. The extention rod shall contain a well-insulated electrical conductor to provide this electrical connection.

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3.2.4 Window. The distal end of the window shall have an external taper to mate with the taper in the light carrier and window attachment, and provide an airtight fit. A clear glass window shall be securely cemented in the proximal end in an airtight manner.

3.2.5 Electrical connector cord. A 72-inch minimum overall length, two-conductor electrical connector cord shall be provided, one end of which shall have a female bayonet-type connector to fit the connector on the light carrier window attachment and the other end of which shall have two jack plugs, one for each conductor.

3.2.6 Lamps and sleeve. Six lamps and a magnifier sleeve for the proximal lamps shall be provided with each instrument. Three of the lamps shall be for distal lighting and three of the lamps shall be for proximal lighting (see 6.1). The magnifier sleeve shall consist of a cylinder to fit the proximal lamp, and shall be provided with a magnifying lens at its distal end.

3.2.7 Insufflator and connector. The insufflator shall consist of an inflation bulb connected through a suitable connector and valve to a net-covered rubber bag which, in turn, shall be cemented to a length of rubber tubing. The distal end of the rubber tubing shall be provided with a metal connector, which shall have a taper on its distal end to mate with the female taper of the insufflator attachment on the light carrier and window attachment, and provide an airtight connection. The entire insufflator shall be airtight.

3.2.8 Rubber parts.

3.2.8.1 Accelerated aging. All rubber parts shall be subjected to the accelerated aging test specified in 4.4.1 and, upon being so subjected, shall shown no stiffening, softening, cracking, or other changes.

3.2.8.2 Breaking strength and ultimate elongation. The breaking strength and ultimate elongation of the tubing and inflation bag shall not decrease more than 25 percent after being subjected to the aging test specified in 4.4.1. The breaking strength and ultimate elongation shall be determined as specified in 4.4.2.

3.2.9 Case. A plush-lined leatherette-covered carrying case shall be provided. The case shall be constructed with interior fittings to retain securely the instrument and components, and to prevent shifting or movement of the contents. The case shall be fitted with a piano hinge or with at least three separate hinges and shall be provided with one or two latches to prevent accidental opening. The case shall be marked with the letters "U" and "S" and the stock number. The letters and numbers shall be approximately 1/2 inch high, and shall be in gold.

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3.3 Airtightness. The sigmoidoscope shall be airtight when tested as specified in 4.4.3.

3.4 Identification marking. The sigmoidoscope shall be clearly and legibly marked with the manufacturer's name or registered trademark.

3.5 Service data. The contractor shall enclose, with each unit, two copies each of one or more booklets which contain complete instructions for installation, operation, maintenance, and a list of component parts. Wiring diagrams shall be included for electrical items or components. Each part of the unit other than common fasteners, etc., shall be identified in the service data and parts list by the manufacturer's part number. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by the Defense Medical Supply Center. The preliminary service data shall be submitted sufficiently early to permit adequate review by the Defense Medical Supply Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order. In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution.

3.6 Workmanship. Workmanship shall be first class throughout. Proctosigmoidoscopes shall be free of imperfections which may detract from their appearance or impair their serviceability. All tapered connections shall assemble and disassemble easily, and shall provide airtight fit when assembled. All electrical parts shall be properly insulated to avoid the possibility of short circuiting, and all electrical connectors shall provide good contact.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in the following table:

Component	Characteristic	Requirement	Test procedure	Unit of product	Inspection level	AQL (percent defective)
Metal parts	Thickness of coating	GG-I-526	GG-I-526	1 unit	S-1	*
	Salt-spray test	GG-I-526	GG-I-526	1 unit	S-1	*
	Boil test	GG-I-526	GG-I-526	1 unit	S-1	*
Inflation bulb	Accelerated aging	3.2.8.1	4.4.1	--	**	**
End item	Airtightness	3.3	4.4.3	1 unit	S-4	1.5
Tubing and inflation bag	Accelerated aging, breaking strength, and ultimate elongation	3.2.8.1 and 3.2.8.2	4.4.1 and 4.4.2	--	**	**

* Acceptance number shall be zero for all sample sizes.
** Each lot manufactured from the same batch of rubber shall be tested as specified in the applicable test procedure. Failure to comply shall be cause for rejection of the lot represented.

4.2.3 Raw materials. Four ounces of each lot of each metal used in the construction of the sigmoidoscopes shall be selected for chemical analysis.

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4.2.4 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The sigmoidoscopes shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Accelerated aging test. Rubber parts shall be tested as specified in method 7221 of Federal Test Method Standard No. 601.

4.4.2 Breaking strength and ultimate elongation. The breaking strength and ultimate elongation shall be determined in accordance with method 4111 or 4121 of Federal Test Method Standard No. 601. In using method 4111 to determine the breaking strength, the value defined as "breaking force" will be used as the value for breaking strength.

4.4.2.1 Specimens. For ultimate elongation and breaking strength tests, straight specimens shall be used.

4.4.3 Airtightness test. The proctosigmoidoscope shall be assembled as for examination (i.e., the proctosigmoidoscope tube, the light carrier and window attachment, the window, the electrical connector cord, and the insufflator with connector) and the distal end of the tube shall be tightly stoppered. Air pressure shall be introduced into the assembled instrument by fully inflating the net-covered rubber bag with the rubber bulb of the insufflator. The proximal end of the proctosigmoidoscope shall be immersed in water to the 25-centimeter division on the tube and after 30 seconds of immersion, there shall be no evidence of air bubbles from the following connections:

(a) The proximal end of the tube with the distal end of the light carrier and window attachment.

(b) The proximal end of the light carrier and window attachment with the distal end of the window.

(c) The female bayonet-type electrical cord connector with the male connector of the light carrier and window attachment.

(d) The male connector of the insufflator with the female connector on the light carrier and window attachment.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) sigmoidoscope, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table I. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	12 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in a carton of appropriate size constructed in accordance with PPP-B-636. Closure shall be adequate to prevent spillage of contents under normal handling. Each unit shall be further packaged in accordance with method IA-8 of MIL-P-116.

5.1.4 Level B.

5.1.4.1 Unit package. Each unit shall be packaged as specified in 5.1.3.1, except that the barrier bag shall not be required.

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5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix of PPP-B-636. In addition, waterproofing shal be as specified in 30.4.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic. Method I closure specified in appendix of PPP-B-636 shall be utilized for fiberboard boxes.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 The lamps for distal lighting referred to in 3.2.6 are Federal stock number 6240-797-0570, LAMP INCANDESCENT, 2.5 volt, 0.3 ampere, Bulb T-3/4, Clear and the lamps for proximal lighting are Federal stock number 6240-797-0560, LAMP, INCANDESCENT, 2.5 volt, 0.3 ampere, Bulb T-1-3/4, Clear.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency -DM

Review activities:
Army - MD
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